

working on an isolated and dedicated digital earth. This system does not offer common-mode noise rejection as needed. This kind of scheme is popularly promoted in the market as a transformerless UPS. But this system does not offer foolproof safety against various line vagaries.

In Fig. 2, the UPS has an isolation transformer built-in, which offers considerable common-mode noise rejection. It provides for using a dedicated earth for computer network. Expect neutral-to-earth voltage of less than 2V, and safety against common hazards like lightning.

Now the next question obviously is what should be the quality of this isolation transformer? Where should this transformer be connected, at UPS input or output? And what more?

An isolation transformer must offer common-mode noise rejection of over 120 dB and inter-winding capacitance of a few picofarads, offering foolproof protection against fast rising transient in the frequency band of over 10 kHz. A super-isolation transformer can protect a computer network even if the primary of the transformer receives common-mode voltage of over 1000,000 volts (as in the event of lightning). In a super-isolation transformer, primary and secondary are separately wound on a split bobbin, reducing its inter-winding capacitance to a few picofarads, making it an excellent noise filter.

It may not be economical to install isolation transformers in high-capacity UPS systems. For example, for a 500kVA UPS, the cost of isolation transformer and space occupied by it can be prohibitive. For such high-capacity systems, the UPS systems have to rely on robust and sound safety from the distribution transformer, which again must have a dedicated earth for neutral and a dedicated earth for digital equipment. But for all moderate capacities, in between, use of super-isolation transformers should not be compromised. Proper lightning and surge protection must be provided at

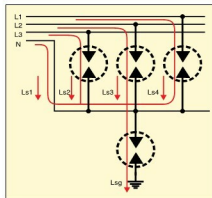


Fig. 3: Typical arrangement of Class B surge arresters for three-phase circuit

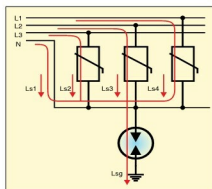


Fig. 4: Arrester with N-PE residual current spark gap as 3+1 circuit

the UPS input. Servers and other delicate electronics is also recommended to be used separately with super-isolation transformers.

Now if you want to include UPS in your protection plan, use proper capacity of the isolation transformer at the input of the UPS. Else, use the same at the UPS output.

Class B and Class C surge arresters—reliable and low-cost defence

Class B surge arresters are specially designed to tackle lightning surges. A single-phase solution uses a set of two surge arresters, one between phase and neutral and the other between neutral and earth.

Similarly, Class C surge arresters protect against fast-rising transients. Upon lightning a Class B surge arrester absorbs all the energy induced by lightning and provides a short circuit path tripping the MCBs connected

at the input supply and eventually protecting the system. Various types of surge arresters are available which offer clamping voltages between 900V and 4000V and absorb induced currents up to 100,000 amps.

Figs 3 and 4 show configuration of Class B and Class C surge arresters in three-phase power circuits. Class B surge arresters work as a spark gap device, having clamping voltage of 900V to 4000V. When lightning strikes, a huge amount of electrical energy may get coupled to the electric power lines. Specially designed spark gap in Class B surge arresters starts conducting and allows several kilo-amps of induced current to flow to earth as shown in Fig. 4.

Class C surge arresters are basically made of metal-oxide varistors having clamping level depending upon industrial, domestic or telecom application. Class B and Class C surge arresters have different work zones. A Class C surge arrester cannot substitute for Class B and vice versa. Class C surge arresters are supposed to be used closer to the user devices, while Class B devices are used at the input line of a facility. If properly installed, surge arresters tend to switch off power supply to the connected circuits through input MCBs or fuses when hit by a surge voltage, thus protecting connected devices.

Earth is the mother of all electrical protections. A dedicated earth for IT equipment is mandatory. There should actually be a pair of earth connections (pits) having earth resistance of less than one ohm. These should be capable of carrying surge currents of several kilo-amperes. Using a copper bus-bar of proper thickness and maintaining low earth resistance in all seasons is mandatory. No computer network can be safe without a proper earth connection. Provision of earth is a challenging task in hilly areas, but its utility cannot be compromised. ●

The author is a director at Everon Energy Systems, New Delhi